The Ozaki Scheme

Emulating FP64 matrix multiplication on low-precision hardware

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Intro

FP64 on new GPUs

GPU	Year	FP64 Vector (TFLOP/s)	FP64 Tensor Core (TFLOP/s)
V100	2017	7.8	—
A100	2020	9.7	19.5
H100	2022	34	67
B200	2024	37	37
B300	**2025**	**1.3**	**1.3**
Rubin	2026	33	200 (emulated)

“The floating-point capability of the platform is not improved – not improved – over the previous generation.” — Jack Dongarra, SC25 (Woodie 2026b)

Emulation: Use the abundant low-precision hardware to get results equivalent to native FP64. The method is the Ozaki scheme (Ozaki et al. 2012).

Various perspectives

People who care a lot about FP64 don’t agree on what emulation is for.

> “FP8 is All You Need: Debunking Hardware FP64 as the HPC Holy Grail”
>
> Satoshi Matsuoka, RIKEN (Eadline 2026)

> “We don’t view this [as a] substitution. These are complementary.”
>
> Darío Gil, U.S. Department of Energy (Woodie 2026b)

> “If there’s a case that this emulation is producing scientifically acceptable results, then suddenly our FP64 cost is now divided by 2X to 3X without actually changing the chip.”
>
> Dan Stanzione, TACC (Prickett Morgan 2026)

> “If you get to wrong answers really fast, that’s not helping anybody. We want to get to right answers fast.”
>
> Joseph George, AMD (Woodie 2026a)

Goal: Trace the development of the Ozaki scheme, and consider what each work tells us about when it can or should be used.

Foundations

Floating-point basics

$$\text{value} = \text{mantissa} \times 2^{\text{exponent}}$$

$$\text{e.g.} \quad 28.016 = 1.751 \times 2^4$$

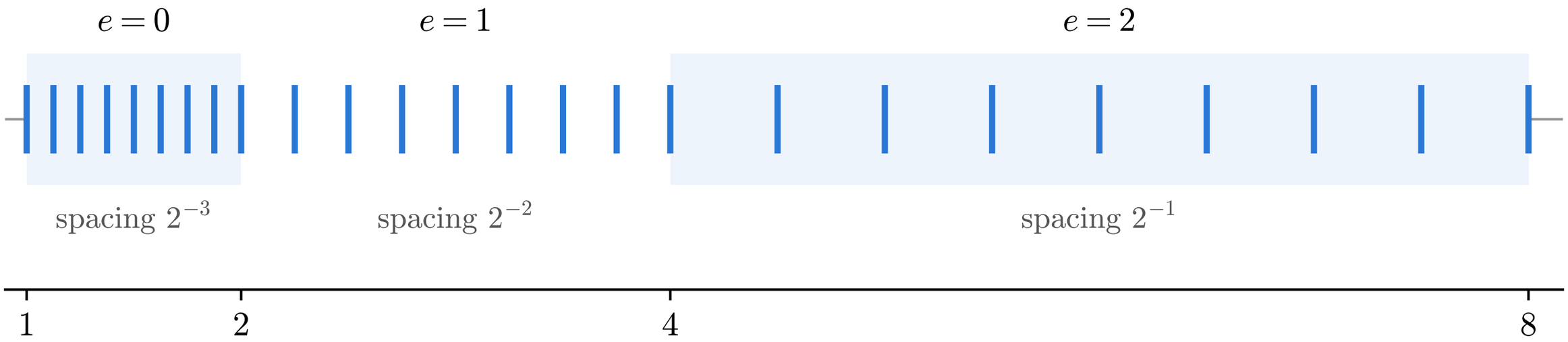
Our discussion is concerned with FP64:

- 53-bit mantissa
- 11-bit exponent

Max relative error from rounding a real number to FP64:

$$2^{-53} \approx 1.11 \times 10^{-16}$$

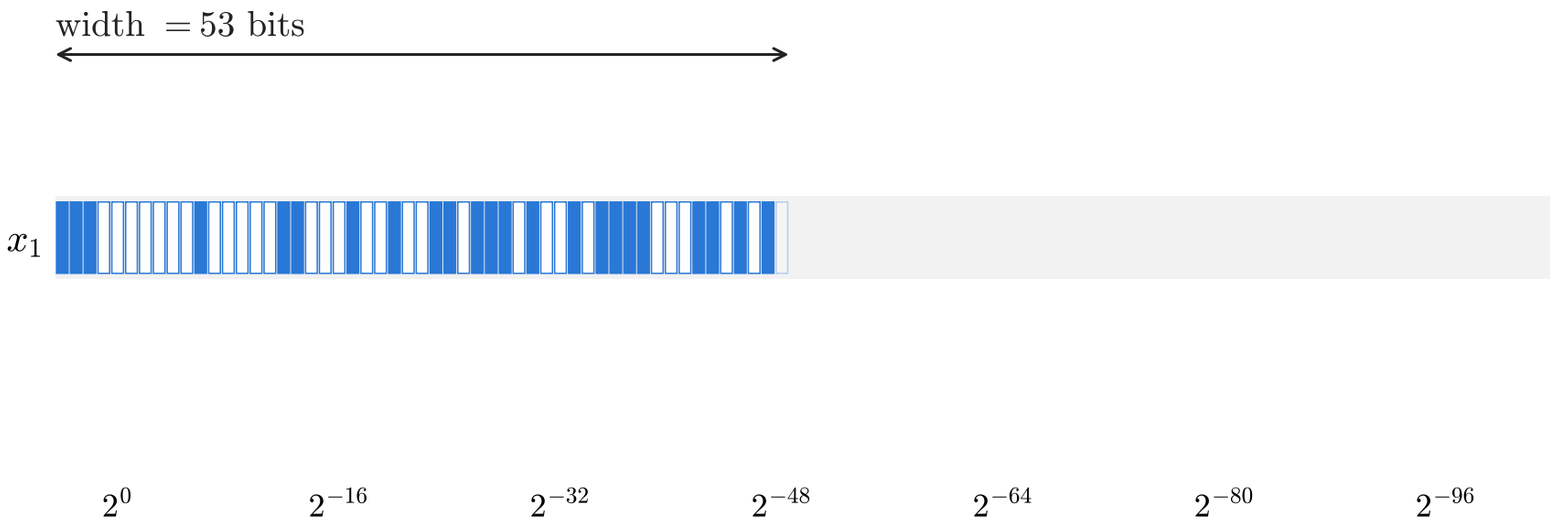
Floating-point space



- **Floating-point space:** Floating-point values are a subset of points on the real number line.
- Density of coverage decreases as magnitude grows

Mantissa space: One value

$$x_1 = 1.751 \times 2^4$$



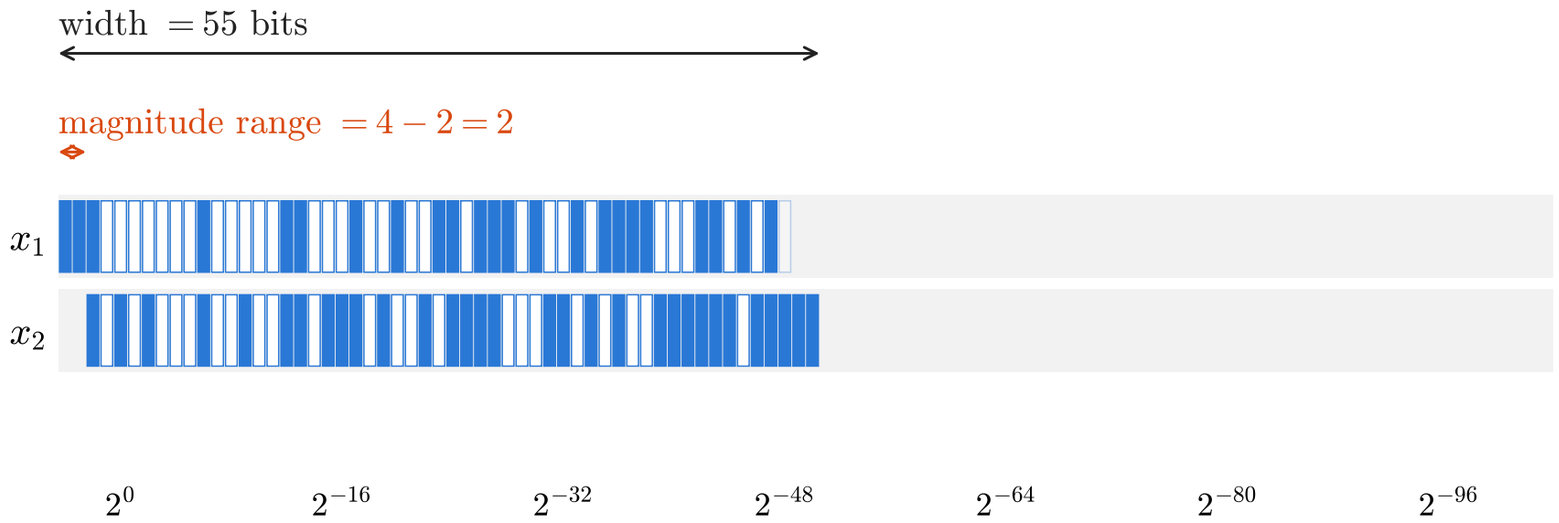
Mantissa space: Visualization of floating-point mantissas, showing the magnitude of each bit

One floating-point value: Mantissa space is 53 bits wide.

Mantissa space: Two values

$$x_1 = 1.751 \times 2^4$$

$$x_2 = 1.317 \times 2^2$$



Width: Distance from the most significant bit of the highest-magnitude value to the least significant bit of the lowest-magnitude value.

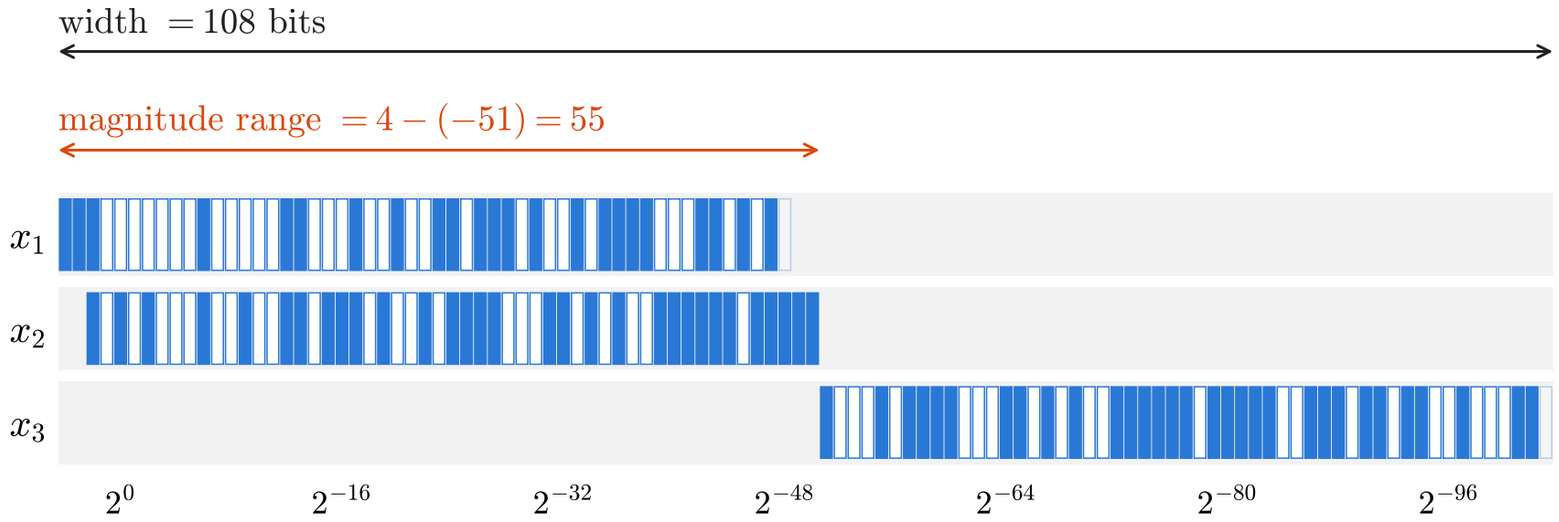
Magnitude range: Difference in exponents of the highest- and lowest-magnitude values.

Mantissa space: Three values

$$x_1 = 1.751 \times 2^4$$

$$x_2 = 1.317 \times 2^2$$

$$x_3 = 1.092 \times 2^{-51}$$



x_1 and x_2: Magnitude range = 2, mantissa space width = 55

x_1, x_2, and x_3: Magnitude range = 55, mantissa space width = 108

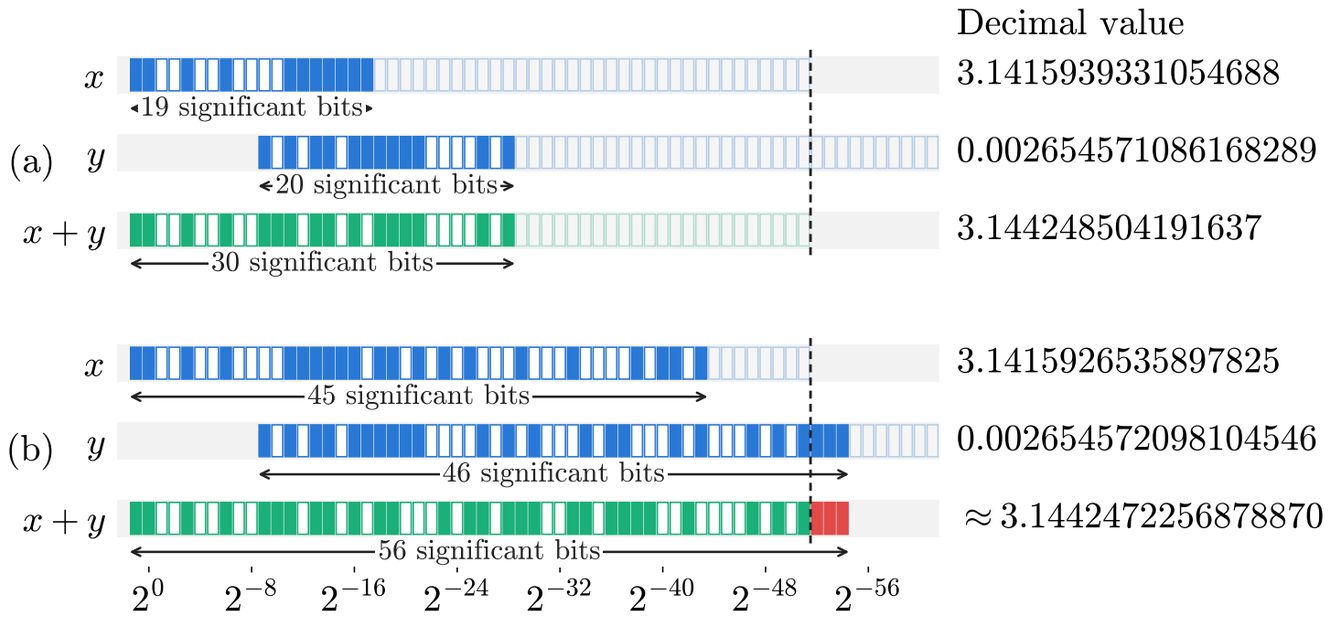
Larger magnitude range → Wider mantissa space

The Ozaki Scheme

Method · Trade-off · Purpose

Rounding in mantissa space

Method → Trade-off → Purpose



The result is rounded only if it doesn’t fit in the 53-bit mantissa.

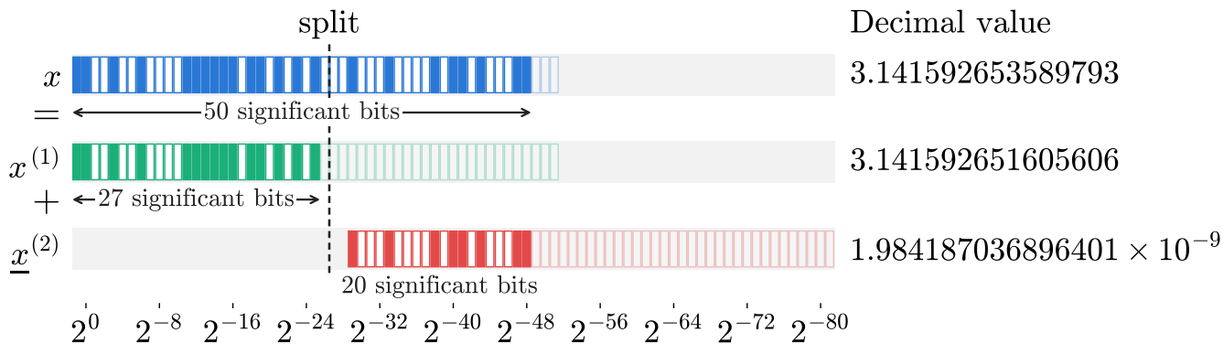
If the operands have few enough significant bits, the result is exact.

Splitting

Method → Trade-off → Purpose

Rump et al. split a single value x into a high part $x^{(1)}$ and a low part $\underline{x}^{(2)}$ (Rump et al. 2008).

$$x^{(1)} = \mathrm{fl}\Big((\sigma + x) - \sigma\Big) \qquad \underline{x}^{(2)} = \mathrm{fl}(x - x^{(1)})$$

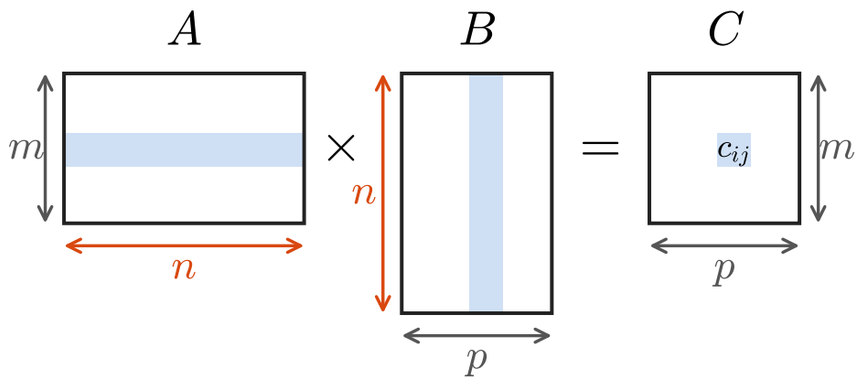


Splitting factor: σ, a power of 2, parameterized by M

Property: $x^{(1)}$ has at most $53 - M$ significant bits

Decomposition of a product

Method → Trade-off → Purpose

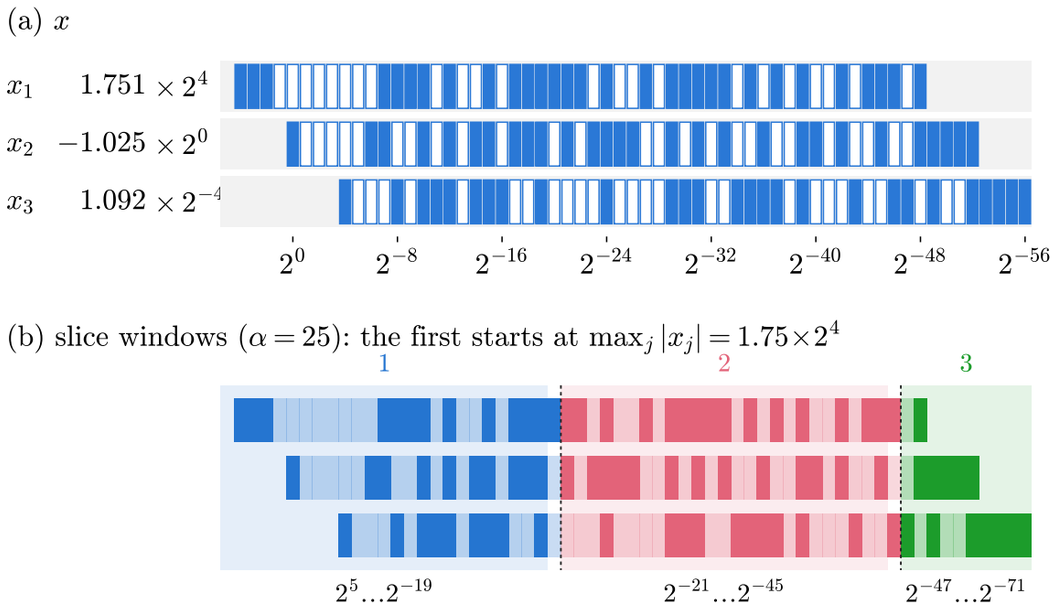


The elements of a matrix product with inner dimension n are dot products between vectors of length n.

Each dot product has n multiplications and $n - 1$ additions, where floating-point error can accumulate.

Splitting a vector

Method → Trade-off → Purpose



FP64 slices

$$
\begin{array}{cccc}
 & x^{(1)} & & \\
 & 1.75\times2^{4} & -1.02\times2^{0} & 1.09\times2^{-4} \\
 & x^{(2)} & & \\
+ & -1.41\times2^{-23} & 1.05\times2^{-22} & 1.06\times2^{-24} \\
 & x^{(3)} & & \\
+ & 1.00\times2^{-48} & -1.94\times2^{-48} & -1.38\times2^{-48}
\end{array}
$$

Scale σ to the largest element of x, then split each remainder again.

Result: x is a sum of slices, each with at most $53 - M$ significant bits

Exact slice products

Method → Trade-off → Purpose

Split y the same way, and the dot product becomes a sum of slice products:

$$x^T y = \left(x^{(1)} + \cdots + x^{(n_x)}\right)^T \left(y^{(1)} + \cdots + y^{(n_y)}\right) = \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \underbrace{x^{(i)^T} y^{(j)}}_{\text{exact}}$$

Rounding error only accumulates when the slice products are added together.

$x^T y$ is more accurate than if it had been computed directly.

The Ozaki scheme

Method → Trade-off → Purpose

1. **Decompose:** Split A and B into slices, fully or partially
2. **Multiply:** Slices are standard FP64 matrices, so slice products use standard FP64 matrix multiplication routines
3. **Accumulate:** Accept rounding error in the sum, or use an accurate summation technique

Trade-off: Accuracy vs. performance

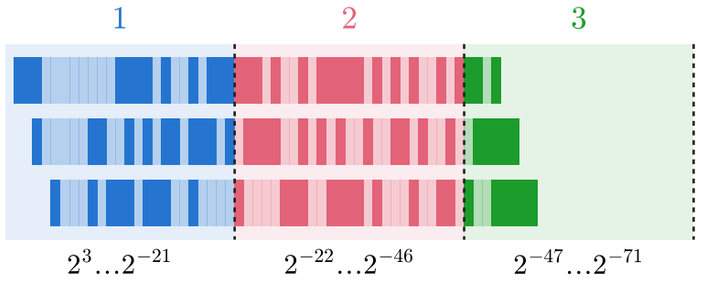
Method → **Trade-off** → Purpose

(a) magnitude range = 4

$x_1 \quad 1.751 \times 2^{2}$

$x_2 \quad -1.025 \times 2^{0}$

$x_3 \quad 1.092 \times 2^{-2}$

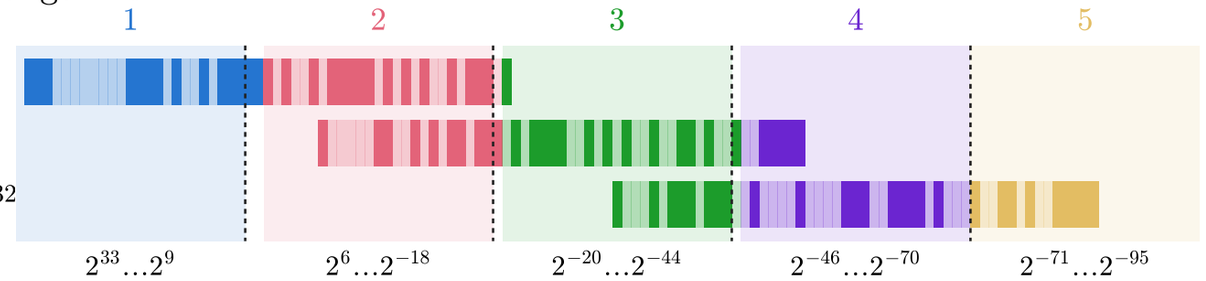


(b) magnitude range = 64

$x_1 \quad 1.751 \times 2^{32}$

$x_2 \quad -1.025 \times 2^{0}$

$x_3 \quad 1.092 \times 2^{-32}$



Slices should cover the whole mantissa space. Any part that isn't covered is represented inexactly.

Wider mantissa space requires more slices to cover it, and more slices means more memory overhead and more computation.

Test matrices

Method → **Trade-off** → Purpose

Ozaki et al. generate test matrices with a controlled magnitude range:

$$a_{ij} = \underbrace{(\mathrm{rand} - 0.5)}_{\text{uniform mantissa}} \times \underbrace{e^{\phi \cdot \mathrm{randn}}}_{\text{normal exponent}}$$

ϕ is the parameter controlling magnitude range.

We'll keep seeing matrices like these in the later evaluations.

The trade-off in the results

Method → **Trade-off** → Purpose

M1 FP64 · **M2** `lssresidual` (INTLAB) · **M3–M5** Ozaki scheme, 2, 3, 4 slices · **M6** XBLAS

Accuracy (maximum componentwise relative error)

ϕ	M_1	M_2	M_3	M_4	M_5	M_6
1	5.64e–010	8.12e–011	7.95e–015	2.20e–016	3.27e–016	1.11e–016
5	3.48e–011	2.39e–011	7.28e–012	2.19e–016	3.24e–016	1.11e–016
10	2.90e–011	2.68e–011	8.88e–011	1.59e–012	2.21e–014	1.11e–016
15	6.81e–012	5.29e–012	5.39e–012	5.60e–012	4.18e–012	1.11e–016

Time relative to standard FP64

ϕ	M_1	M_2	M_3	M_4	M_5	M_6
1	1	3.30	3.34	7.41	10.4	99.1
5	1	3.29	3.35	7.62	11.7	95.9
10	1	3.30	2.40	3.41	8.60	84.9
15	1	3.30	2.74	3.81	7.13	97.1

As ϕ increases, the schemes with lower slice counts see more accuracy degradation. More slices take longer.

What it was for

Method → Trade-off → **Purpose**

At this point, the Ozaki scheme is only meant to be an efficient and more accurate way of performing matrix multiplication.

Tested on: Intel Core 2 Duo and Core 2 Extreme, MATLAB, $n = 1000$ and 2000

Result: Accuracy near XBLAS at low ϕ, in a much smaller fraction of the time (8–35× faster)

The evaluations show the Ozaki scheme can be more efficient than other methods for accurate matrix multiplication, though not over a wide range of cases.

A wrapper around matrix multiplication

Method → Trade-off → **Purpose**

Slices are regular FP64 matrices, so most of the work can be offloaded to optimized BLAS routines.

The Ozaki scheme is essentially a wrapper around DGEMM, transforming one DGEMM call into several.

This also makes it promising as a method that could be integrated transparently into existing libraries.

Takeaways: Ozaki et al. (2012)

Role: A faster way of performing accurate matrix multiplication

Trade-off: Magnitude range determines the slice count, which trades accuracy against performance and memory

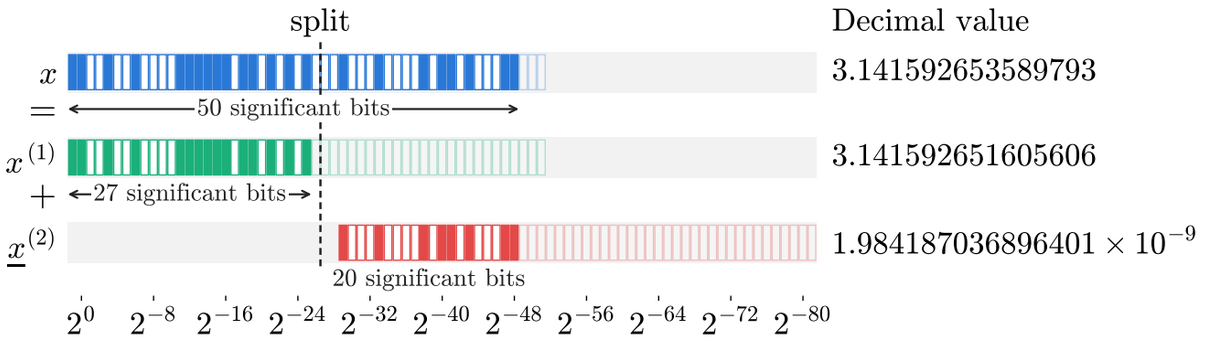
Ozaki on IMMU

Tensor Cores · Integer Slices · New Possibilities

Ozaki is shaped like Tensor Cores

Tensor Cores → Integer Slices → New Possibilities

Tensor Cores (2018): Dedicated mixed-precision matrix multiplication circuits. Inputs can be in lower precision than outputs.



FP64 slices are deliberately restricted to a fraction of the mantissa (27 of 53 bits here). That headroom exists only so the *result* can be exact in FP64.

There's no reason the slices themselves can't use smaller formats.

The first Ozaki scheme on Tensor Cores (2020) used FP16-FP32 cores (Mukunoki et al. 2020)

Ootomo et al. (2024)

Tensor Cores → Integer Slices → New Possibilities

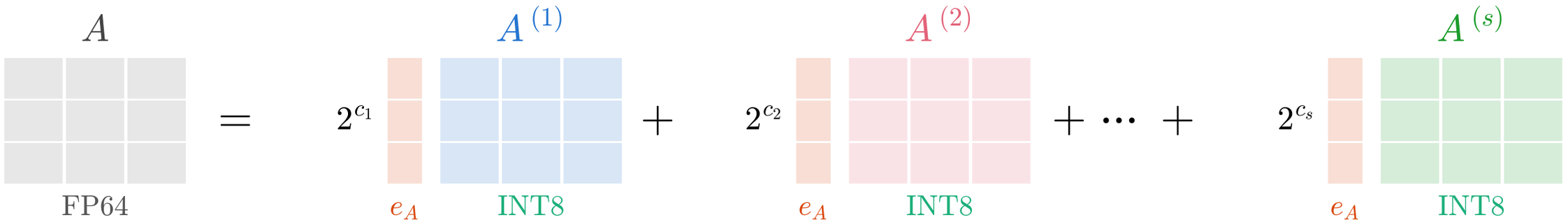
DGEMM on integer matrix multiplication unit

Hiroyuki Ootomo, Katsuhisa Ozaki, Rio Yokota · *The International Journal of High Performance Computing Applications*, 2024 (Ootomo et al. 2024)

The Ozaki scheme on integer Tensor Cores (INT8)

Integer slices

Tensor Cores → **Integer Slices** → New Possibilities

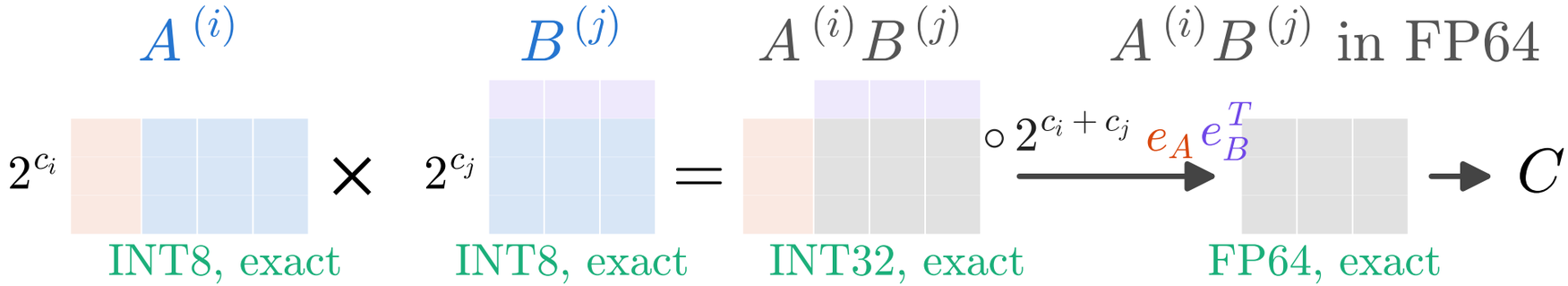


Ootomo et al. use a block-float format: Each slice contains integer values aligned to a shared exponent.

Exponents are divided out and stored in separate vectors.

The IMMU Ozaki scheme

Tensor Cores → **Integer Slices** → New Possibilities



1. **Decompose:** Split into INT8 slices and scaling vectors
2. **Multiply:** Compute slice products on integer Tensor Cores, then convert products back to FP64 using the shared exponents and scaling vectors
3. **Accumulate:** Sum the FP64 products

What the slices leave out

Tensor Cores → **Integer Slices** → New Possibilities

With s INT8 slices, the product is only approximated:

$$AB \approx \sum_{i+j \leq s+1} A^{(i)} B^{(j)}$$

Bits below the last slice are cut off, and products with $i + j > s + 1$ are skipped.

The main error comes from insufficient splitting, not from rounding.

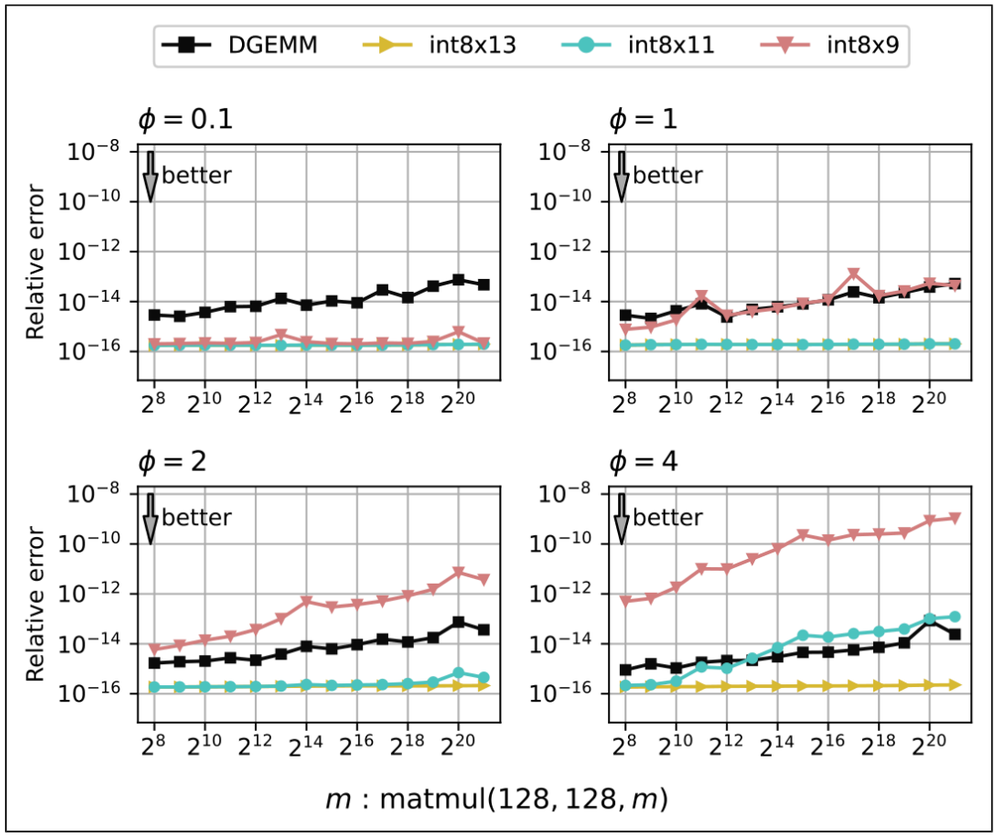
Experimental setting

Tensor Cores → Integer Slices → **New Possibilities**

- **Inputs:** Same synthetic matrices as Ozaki et al. (2012), $\phi = 0.1, 1, 2, 4$
- **Hardware:** A100, TITAN RTX, RTX A6000, RTX 6000 Ada
- **Error:** Maximum componentwise relative error, against a double-double reference
- **Variants:** INT8x9, INT8x11, INT8x13 (9, 11, 13 slices)

Results: Accuracy

Tensor Cores → Integer Slices → **New Possibilities**



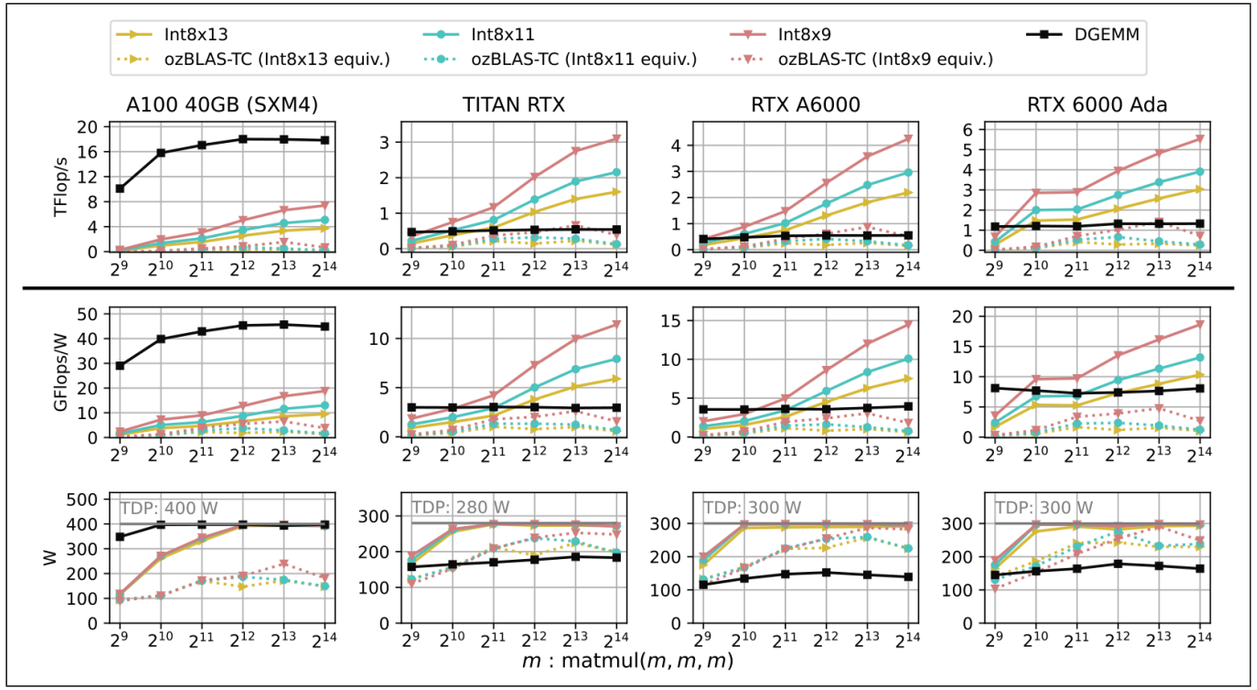
At low ϕ, all variants stay near 10^{-16}.

As ϕ grows, variants with fewer slices degrade first.

We see again that lower slice counts improve performance at the cost of accuracy.

Results: Throughput

Tensor Cores → Integer Slices → **New Possibilities**



A100: At best 8 TFlop/s, vs. DGEMM's ~17.

Consumer GPUs: Up to 3, 4, 6 TFlop/s, vs. DGEMM's 0.5–1.

Speedup depends on the hardware. The new implementation let's us take proper advantage of what the hardware provides.

FP64 : INT8 peak: **A100** 1 : 32 · **TITAN RTX** 1 : 522 · **RTX A6000** 1 : 515 · **RTX 6000 Ada** 1 : 520

New possibilities

Tensor Cores → Integer Slices → **New Possibilities**

On some hardware, the Ozaki scheme can be used not just to produce a more accurate matrix product, but to produce it *faster* than the standard DGEMM routine, and more power efficiently.

But: Testing is still limited to synthetic matrices, in a narrow range of sizes. Speedup only shows up for *large enough* matrices, and memory overhead is not measured.

More slices, smaller slices

Tensor Cores → Integer Slices → **New Possibilities**

Memory: INT8 slices hold at most 8 bits, so we need more of them. But one FP64 slice takes the memory of 8 INT8 slices, so we can often come out ahead.

Compute: Where INT8 throughput is high enough, we can compute some number of INT8 slice products faster than one FP64 matrix multiplication.

Usability

Tensor Cores → Integer Slices → **New Possibilities**

Users have to manually tune the slice count, based on:

- **Desired accuracy:** Can be difficult to know
- **Magnitude range of their matrices:** Easy to measure, but costly at small sizes
- **Their hardware:** How many slices are equivalent to one FP64 matrix multiplication

Takeaways: Ootomo et al. (2024)

Role: Faster than the standard DGEMM routine, on some hardware

Trade-off: Unchanged, but with more (smaller) slices, and users still tune the slice count manually

Ozaki with ADP

FP64 Emulation · ESC · ADP

Schwarz et al. (2026)

FP64 Emulation → ESC → ADP

Guaranteed DGEMM Accuracy While Using Reduced Precision Tensor Cores Through Extensions of the Ozaki Scheme

Angelika Schwarz, Anton Anders, Cole Brower, Harun Bayraktar, John Gunnels, Kate Clark, RuQing G. Xu, Samuel Rodriguez,

Sebastien Cayrols, Pawel Tabaszewski, Victor Podlozhnyuk NVIDIA · *SCA/HPCAsia '26*, 2026 (Schwarz et al. 2026)

2025: FP64 emulation added to cuBLAS (NVIDIA 2025)

Use emulation when it is beneficial, and standard DGEMM when it is not

A new goal

FP64 Emulation → ESC → ADP

Ozaki et al. (2012)

Goal: As little rounding error as possible

Slice width chosen so slice products are provably exact

Schwarz et al. (2026)

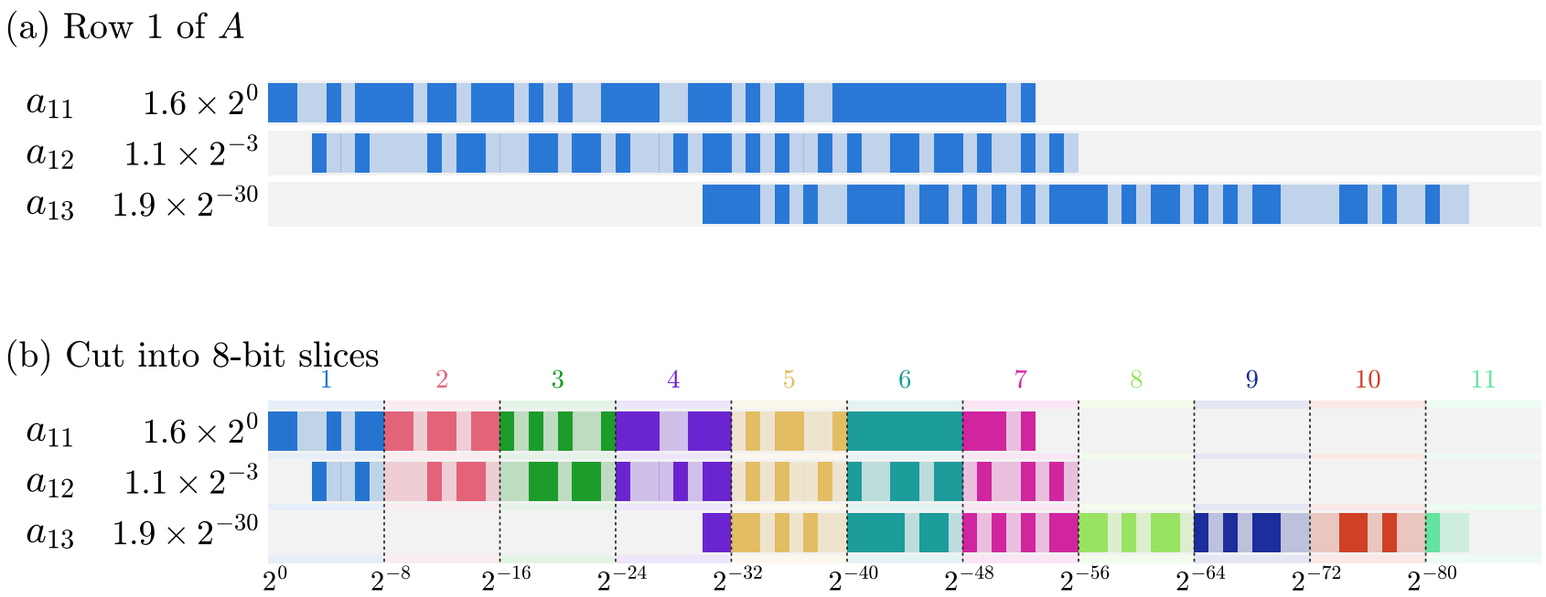
Goal: FP64-equivalent accuracy

Slices always contain eight significant bits

Aiming for FP64-equivalent accuracy gives us a foothold for reasoning about how many slices we need.

Every slice uses all 8 bits

FP64 Emulation → **ESC** → ADP



Slices are cut from the largest value down, eight bits at a time.

How many slices do we actually need?

Which bits matter

FP64 Emulation → **ESC** → ADP

$$
\underset{A}{\begin{bmatrix} 1.6\times 2^{0} & 1.1\times 2^{-3} & 1.9\times 2^{-30} \\ 1.5\times 2^{-1} & 1.6\times 2^{0} & 1.6\times 2^{-2} \\ 1.9\times 2^{0} & 1.5\times 2^{-2} & 1.6\times 2^{-1} \end{bmatrix}} \cdot \underset{B}{\begin{bmatrix} 1.4\times 2^{-20} & 1.4\times 2^{-1} & 1.2\times 2^{0} \\ 1.0\times 2^{-8} & 1.1\times 2^{0} & 1.8\times 2^{-2} \\ 1.5\times 2^{0} & 1.4\times 2^{-2} & 1.0\times 2^{-1} \end{bmatrix}} = \underset{C}{\begin{bmatrix} c_{11} & \bullet & \bullet \\ \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \end{bmatrix}}
$$

largest in the row / column

behind the largest term of c_{11}

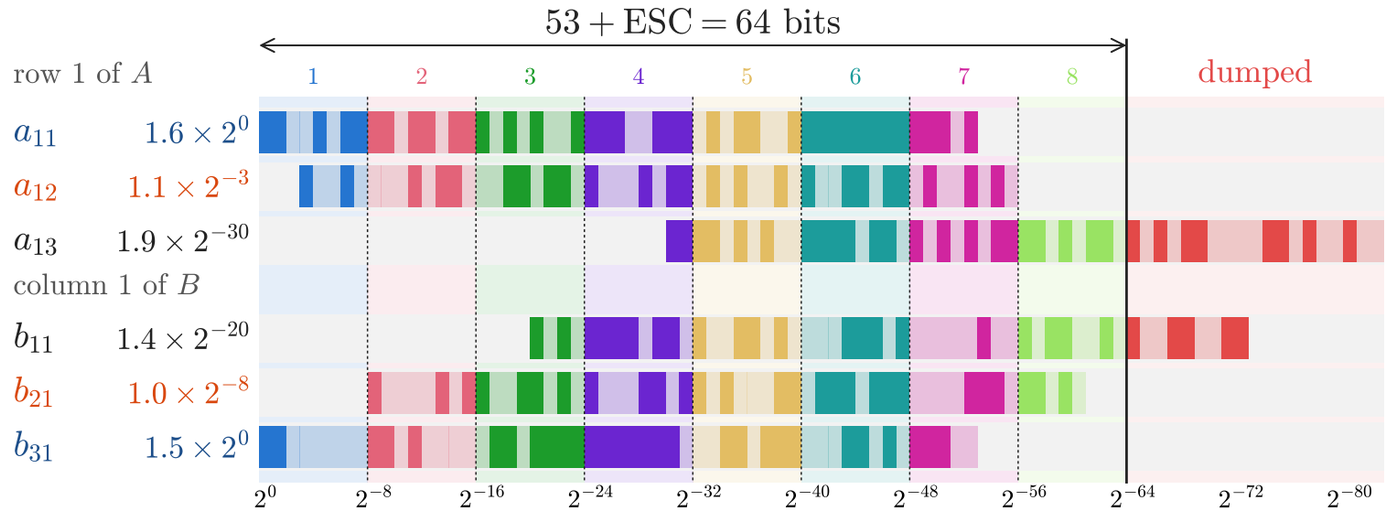
$$
c_{11} = \underbrace{a_{11}b_{11}}_{\approx 2^{-20}} + \underbrace{a_{12}b_{21}}_{\approx 2^{-11}} + \underbrace{a_{13}b_{31}}_{\approx 2^{-30}}
$$

Slices should cover every mantissa bit of the largest values, and of the values behind the largest term.

ESC: $0 + 0 - (-11) = 11$ extra bits

Slices over 53 + ESC bits

FP64 Emulation → **ESC** → ADP

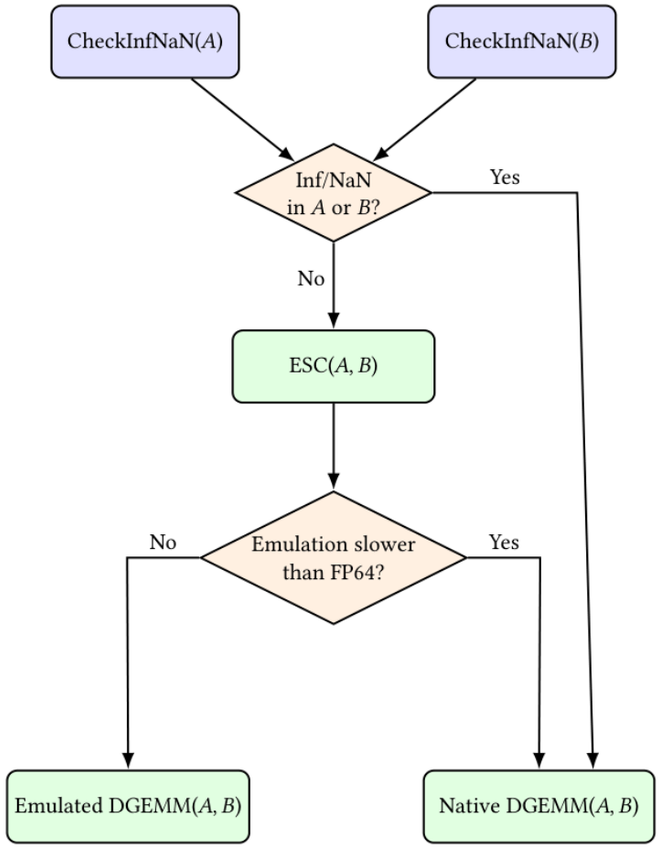


Slices cover $53 + \mathrm{ESC}$ bits. The rest is dumped.

Enough for FP64-equivalent accuracy? Asserted, not proven.

Automatic Dynamic Precision (ADP)

FP64 Emulation → ESC → **ADP**



Check for Inf/NaN, compute ESC, then decide: Emulate, or fall back to native DGEMM

Goal: A matrix product at least as accurate as native FP64, in at least the same amount of time

How the heuristic decides “emulation slower than FP64?” is not described.

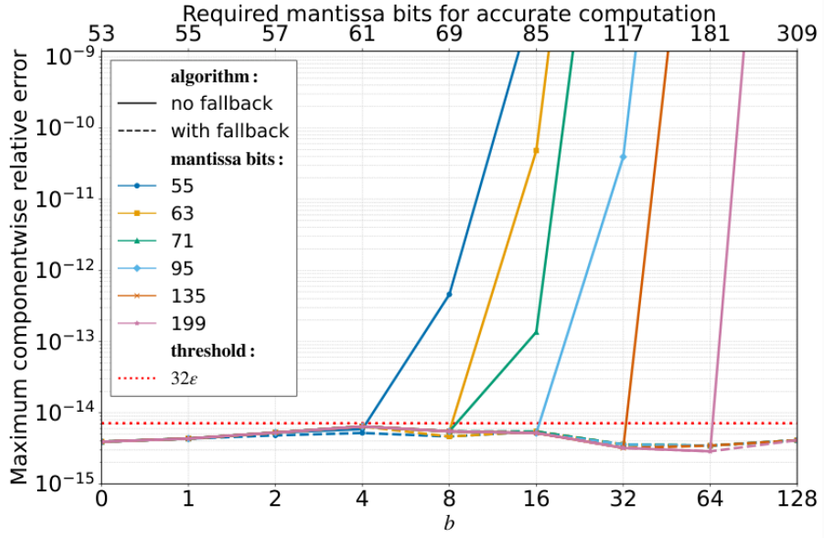
Experimental setting

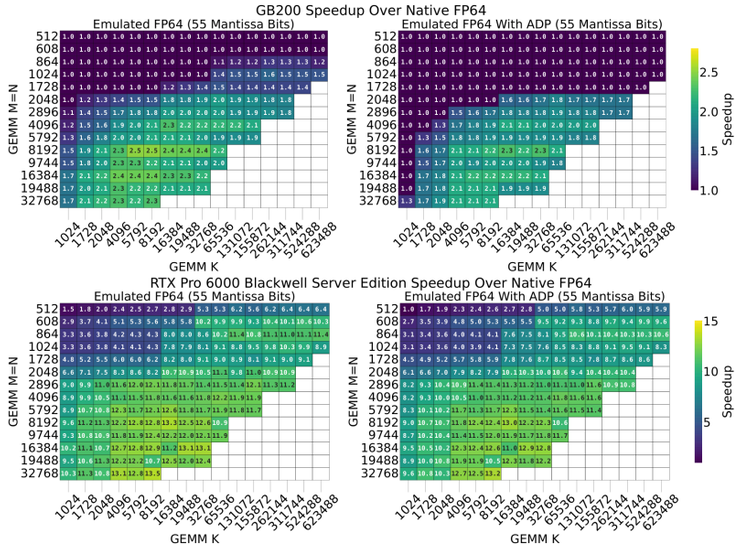
FP64 Emulation → ESC → **ADP**

- **Inputs:** Demmel et al.'s test matrices, with a parameter b controlling exponent range (Demmel et al. 2026)
- **Hardware:** GB200, RTX Pro 6000 Blackwell Server Edition
- **Accuracy:** Maximum componentwise relative error, against a standard FP64 matrix multiplication
- **Variants:** With and without fallback to native DGEMM

Evaluation

FP64 Emulation → ESC → **ADP**





FP64 : INT8 peak: **GB200** 1 : 125 · **RTX Pro 6000** 1 : 512

With fallback: Accuracy indistinguishable from native DGEMM. Up to 2.3× faster on GB200, and 13.2× on RTX Pro 6000.

What the evaluations tell us

FP64 Emulation → ESC → **ADP**

The Ozaki scheme is now an explicit alternative to DGEMM, but it is still evaluated the way it was when it was an accurate matrix multiplication scheme.

But: No adversarial inputs, no look inside the fallback heuristic, and memory overhead is still not measured.

Impacts of implementation

FP64 Emulation → ESC → **ADP**

Accuracy: Slice products are no longer guaranteed exact, though this only matters for very large matrices

Memory: Eight significant bits in every slice is the best memory overhead the scheme can get

Usability: Emulation is invisible to the user… but the decision to emulate is a black box

Takeaways: Schwarz et al. (2026)

Role: FP64 emulation, a drop-in replacement for DGEMM used only when it is beneficial

Trade-off: Handled automatically by ESC and ADP, but how ADP makes that call is not described

Ozaki in Applications

Early Examples · DMRG · Results

Early examples

Until this point, every evaluation has been on synthetic workloads. Are there matrix products in actual applications that would benefit from emulation?

Quantum circuit simulation (Ootomo et al. 2024)

2–4.85× faster than cuBLAS ZGEMM, same or better accuracy

(Random unitary circuits: “toy models”)

QR factorization (Schwarz et al. 2026)

Without ADP: 5× faster, error 2.9e-12

With ADP: 3.5× faster, error 7e-16

Brower et al. (2026)

Early Examples → **DMRG** → Results

Mixed-Precision Ab Initio Tensor Network State Methods Adapted for NVIDIA Blackwell Technology via Emulated FP64 Arithmetic

DMRG

DMRG: Iterative algorithm for finding the lowest-energy state of a molecule

Every iteration: Many DGEMMs, some in rounding-sensitive steps (eigensolver, SVD)

Bond dimension D: Higher is more accurate, but takes more memory and compute

From D to DGEMMs

Early Examples → **DMRG** → Results

The matrices in DMRG are $D \times D$. Symmetries split them into smaller blocks, and the DGEMMs run on the blocks, not on the whole matrix.

DGEMM structure from the group's hybrid CPU-GPU implementation (Menczer and Legeza 2025)

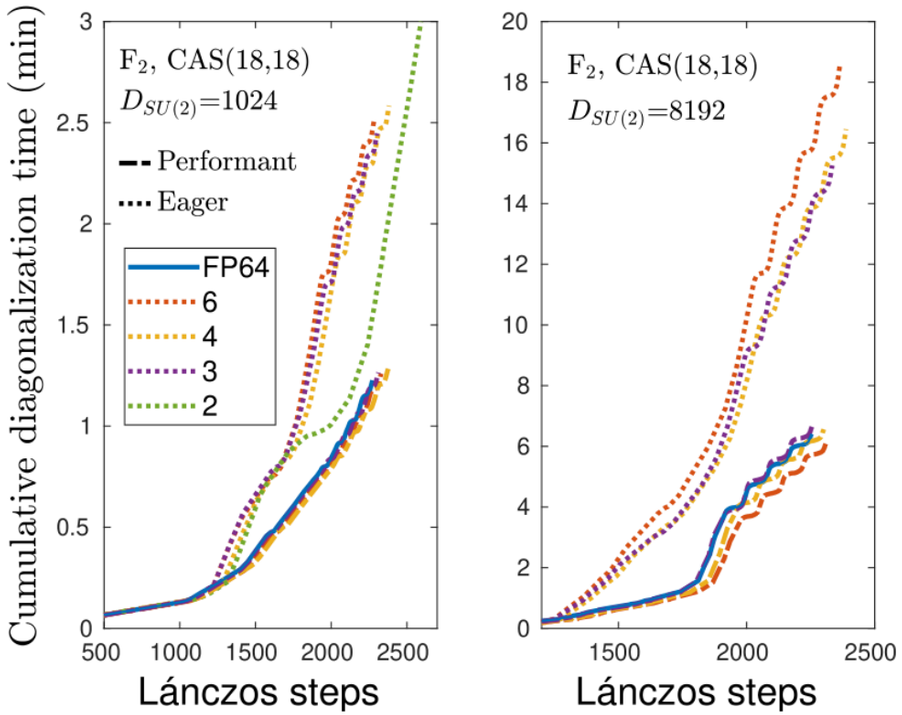
A lot of DGEMM work, but it arrives as many DGEMMs much smaller than D.

Accuracy

Early Examples → DMRG → **Results**

Performance

Early Examples → DMRG → **Results**

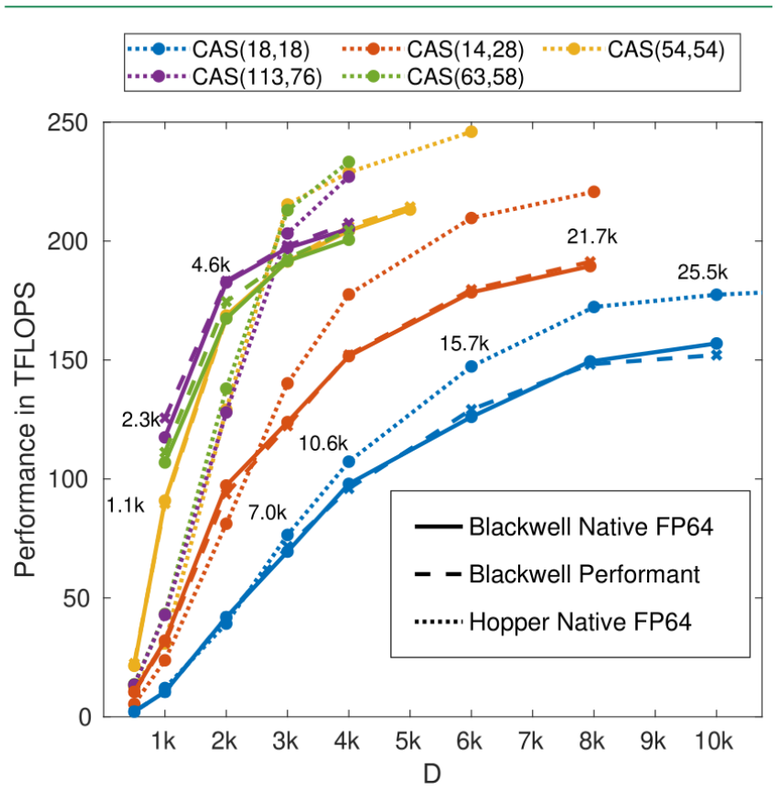


Eager: Emulate every DGEMM. Slower than native FP64, even with 2 slices.

Performant: The library decides. Matches FP64, slightly faster at $D = 8192$.

Performance at scale

Early Examples → DMRG → **Results**



Native FP64 vs. performant emulation on B200, with H100 for reference

Performant mode tracks native FP64. Slight gains only for the largest D and systems.

What the application tells us

Early Examples → DMRG → **Results**

Accuracy: Four to six slices reach chemical accuracy. Two slices break DMRG entirely.

Speed: No real gain yet. Emulation is “often slower for memory-bound matrix multiplications”, and sector DGEMMs are small.

Caveat: Prerelease cuBLAS, and the paper defers performance to a later study.

Takeaways: Brower et al. (2026)

Role: FP64 emulation inside a real scientific code, with the library deciding DGEMM by DGEMM

Trade-off: Accuracy holds with few slices, but speedup depends on DGEMM size, which is set by sector sizes, not D

Discussion

Tracing the Ozaki scheme

Accurate matrix products on CPUs
Ozaki et al. (2012)

→ **Hardware**

Faster than FP64 on integer Tensor Cores
Ootomo et al. (2024)

→ **Usability**

Drop-in FP64 replacement in cuBLAS
Schwarz et al. (2026)

→ **Stability**

FP64 emulation in a scientific application
Brower et al. (2026)

From a method for accurate matrix multiplication that was never expected to compete with FP64 on speed, to a potential FP64-killer

Performance comes from hardware

Integer slices don’t reduce the work: More slices ($3 \rightarrow 9$), more slice products

Speedups follow the hardware: Faster than FP64 at 1 : 512 (consumer GPUs), slower at 1 : 32 (A100)

Hardware is trending in exactly the direction that accelerates the scheme.

NVIDIA and FP64

NVIDIA cuts FP64 throughput: B200 37 → B300 1.3 TFLOP/s

NVIDIA also ships FP64 emulation in cuBLAS, and authors most of its evaluations.

Meanwhile, AMD's MI250X and MI300A (Frontier and El Capitan) keep strong FP64 throughput.

Emulation is most attractive exactly where FP64 has been deprioritized.

What we can conclude

Every speedup: On hardware where FP64 has been deprioritized for AI

Every accuracy test before DMRG: Synthetic matrices, with one magnitude-range parameter

“FP8 is all you need”: The evidence in these works does not support the claim that emulation can largely replace native FP64.

It supports the moderate position: The Ozaki scheme is a useful tool where it is useful.

Future work

- A systematic characterization of how the scheme fails, on structured and real-application matrices
- A clearer picture of memory overhead in full applications
- Automatic slice-count selection below the FP64-equivalent bound: The scheme as a variable-precision method
- Lightweight heuristics for deciding when to emulate

Conclusion

Everything else

- Is INT8 support here to stay?
- Ozaki-II

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